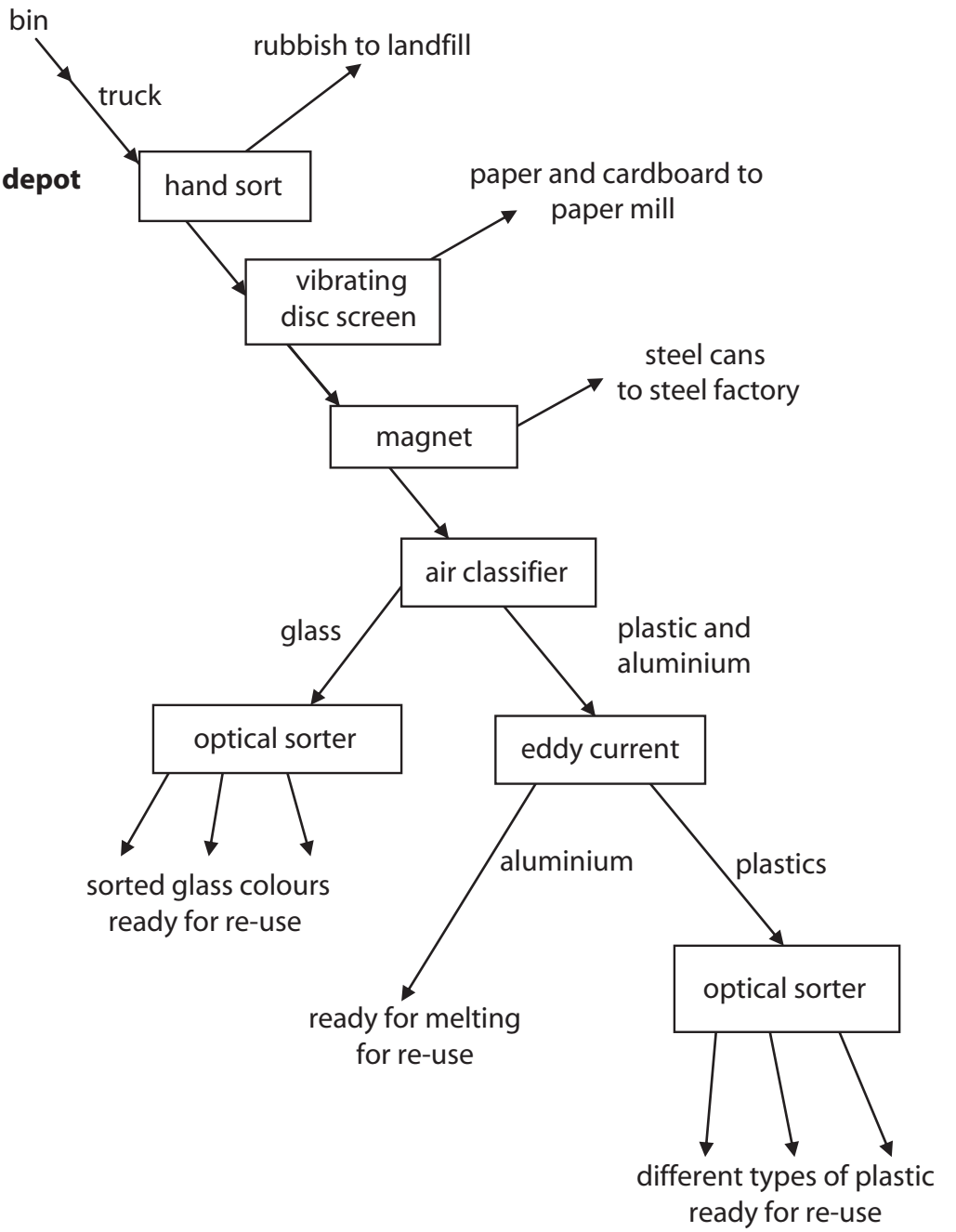
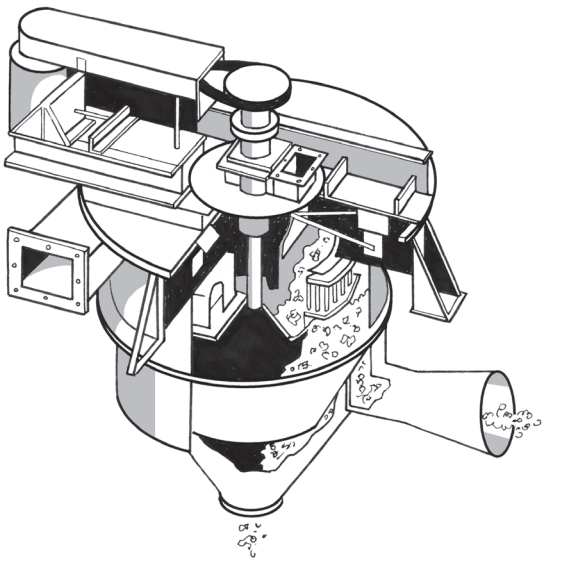
Recycling separators

After your recycling bin is collected the rubbish needs to be sorted before it can be **recycled**. Municipal councils use systems like this for sorting recycled material.



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Air classifier



Look at the Resource sheet 'Recycling separators'. An air classifier separates items by creating strong blasts of air. The air lifts lighter objects such as plastic containers and aluminium cans. Glass jars and bottles are heavier, so they stay on the conveyor belt.

☞ **1** Look carefully at the diagram, then cross out the incorrect words in each sentence.

Glass containers and plastic containers and aluminium cans will leave through the top of the air classifier.

Glass containers and plastic containers and aluminium cans will leave through the bottom of the air classifier.

2 Very small pieces of broken glass are usually removed before the waste reaches the air classifier. Can you explain why this is done?

_____

_____

Density

Although we often talk about lighter and heavier objects, it is more scientific to refer to **dense** and **less dense** objects. Density measures how tightly packed an object's mass is. A pencil and a kitchen sponge weigh about the same, but a pencil is much smaller – the pencil is more dense than the sponge.

☞ **3** Cross out the incorrect item in each of the brackets.

Dense objects have their mass [tightly packed, loosely packed]. An example of a dense object is an empty [glass jar, aluminium can].

Low-density objects have their mass [tightly packed, loosely packed]. An example of a low-density object is a [glass bottle, plastic drink bottle].

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